

CAPE YOUNG, NU, Canada

WMO: 719390

Lat: 68.933N

Lon: 116.917W

Elev: 52

StdP: 14.67

Time Zone: -7.00 (NAM)

Period: 82-93

WBAN: 26114

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-41.7	-38.5	-41.5	0.5	-33.7	-39.3	0.6	-33.0	47.9	-11.7	42.4	2.2	6.0	270	1.081

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	67.6	57.1	61.0	53.7	55.9	50.6	57.2	65.5	53.9	60.8	50.6	55.8	11.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.1	60.1	61.9	49.6	52.7	57.5	46.5	46.8	53.5	24.6	65.3	22.6	60.7	20.6	56.1	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
33.6	28.8	25.0	-44.3	76.8	2.9	5.0	-46.4	80.4	-48.1	83.4	-49.8	86.2	-51.9	89.9		
			-37.4	65.6	1.5	7.5	-38.5	71.0	-39.4	75.4	-40.2	79.6	-41.3	85.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-20.8	-21.1	-15.3	-2.4	18.3	37.1	44.5	43.9	35.1	19.0	-5.9	-14.8
	DBStd	26.83	11.08	13.06	10.83	11.68	10.23	7.18	8.14	6.52	5.52	9.42	12.70	11.26
	HDD50	14692	2196	1991	2025	1572	983	395	217	216	448	962	1677	2010
	HDD65	20088	2661	2411	2490	2022	1448	838	638	654	897	1427	2127	2475
	CDD50	80	0	0	0	0	0	7	46	27	0	0	0	0
	CDD65	2	0	0	0	0	0	0	2	0	0	0	0	0
	CDH74	31	0	0	0	0	0	0	27	4	0	0	0	0
	CDH80	7	0	0	0	0	0	0	7	0	0	0	0	0
Wind	WSAvg	11.2	11.7	11.9	11.5	9.9	9.6	10.0	10.2	11.3	12.6	13.3	10.8	11.2
Precipitation	PrecAvg	7.0	0.2	0.2	0.2	0.3	0.3	0.6	1.3	1.4	1.1	0.7	0.4	0.2
	PrecMax	11.1	0.5	0.6	0.6	1.9	1.1	1.8	3.4	3.1	2.9	2.6	1.3	0.9
	PrecMin	3.3	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.3	0.0	0.0	0.0
	PrecStd	2.0	0.1	0.2	0.2	0.4	0.3	0.5	0.8	0.7	0.6	0.6	0.3	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.3	19.7	17.5	30.0	41.1	69.7	77.1	73.5	55.8	39.6	37.5	21.4
		MCWB	15.6	19.2	16.5	29.4	38.6	56.0	61.8	61.2	49.3	38.2	33.4	20.9
	2%	DB	10.3	12.2	12.2	24.5	36.9	59.6	69.6	64.3	48.5	37.2	25.1	14.0
		MCWB	10.0	11.9	11.8	23.7	35.0	51.4	57.6	55.8	44.7	35.8	24.1	13.6
	5%	DB	1.8	4.7	7.5	19.6	34.1	52.5	63.0	59.4	44.8	33.6	18.2	8.9
		MCWB	1.5	4.3	7.1	19.0	33.0	46.9	54.4	53.4	42.6	32.5	17.5	8.4
	10%	DB	-4.3	-2.1	1.8	15.6	32.2	47.8	57.6	54.0	42.5	31.6	12.3	3.3
		MCWB	-4.5	-2.5	1.4	15.1	31.7	43.7	51.8	49.7	40.8	31.0	11.7	2.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.2	19.5	16.7	29.3	38.6	55.8	64.5	62.4	49.5	38.5	34.0	20.7
		MCDB	16.8	19.9	17.2	29.8	40.8	67.5	76.0	71.2	56.1	39.4	36.8	21.0
	2%	WB	10.6	12.5	12.0	23.6	35.0	51.5	58.5	56.6	45.0	35.7	23.7	13.2
		MCDB	10.8	12.7	12.3	24.2	36.1	59.3	67.4	62.4	47.7	36.8	24.8	13.5
	5%	WB	2.6	6.0	7.3	18.8	33.2	47.3	55.3	53.2	42.8	32.4	17.7	8.5
		MCDB	2.8	6.4	7.8	19.3	34.0	52.5	63.1	58.7	44.5	33.0	18.3	9.0
	10%	WB	-3.5	-1.1	1.9	14.8	31.6	43.4	51.5	50.1	40.7	30.6	11.7	3.0
		MCDB	-3.3	-0.9	2.2	15.2	32.3	47.0	57.4	53.9	42.0	31.1	12.2	3.4
Mean Daily Temperature Range	5% DB	MDBR	11.0	12.2	13.4	15.4	12.0	10.0	12.9	10.3	7.5	9.3	10.9	11.7
		MCDBR	16.1	17.2	16.9	16.8	10.8	21.0	25.6	20.7	12.9	8.2	14.1	13.9
		MCWBR	16.0	16.4	16.8	16.8	9.9	14.1	15.9	13.4	9.9	7.7	13.8	13.9
	5% WB	MCDBR	16.4	16.1	16.9	16.5	10.8	20.3	24.8	19.3	11.7	7.5	14.0	13.6
		MCWBR	16.3	16.0	16.9	16.6	10.0	14.2	16.7	13.4	9.3	7.2	13.8	13.6
Clear-Sky Solar Irradiance	taub	0.160	0.212	0.248	0.294	0.335	0.330	0.352	0.341	0.300	0.243	0.157	N/A	
	taud	1.886	2.047	2.190	2.128	2.115	2.340	2.389	2.464	2.509	2.167	1.921	N/A	
	Ebn at Noon	22	181	251	267	267	275	263	254	236	163	19	0	
	Edh at Noon	2	14	23	34	40	33	30	25	18	12	1	0	
All-Sky Solar Radiation	RadAvg	8	151	668	1433	2017	2017	1669	1094	538	152	22	0	
	RadStd	1	12	31	66	109	141	116	110	60	22	2	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page